

Risk Wording Dominates Retrieved Return Content in a Paired Trading-Agent Replay

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Abstract

Memory is often added to language agents on the premise that retrieved experience changes later decisions. That premise is causal: it requires distinguishing the retrieved facts from the language that introduces them. We conduct three paired prompt experiments in a teacher-forced SPY replay. Experiment 1 re-audits a previously reported negative-memory effect in one local Llama-3 8B build: across 24 setups with matured memories, four direct memory/frame conditions produced zero action switches. Experiment 2 fixes a 19-setup sample and crosses three local models, three retrieval compositions, and five warning severities (855 decisions). At every non-binding severity, negative-only and balanced retrieved returns produced zero paired action switches in all 12 planned model-by-severity comparisons. In contrast, warning severity changed actions sharply: Llama 3.2 3B and Llama 3 8B switched from 0/19 to 19/19 ABORTs under strong caution across every retrieval composition; Qwen 2.5 7B switched at the milder caution level. Experiment 3 removes all memory, historical, and outcome language: generic strong caution alone reproduced 19/19 switches in both Llama models. These results show that, in this bounded replay, risk wording is sufficient to drive observed action suppression while retrieved return valence showed no detected content-specific effect. They do not establish live-trading performance, a general property of LLMs, or the usefulness of memory architectures.

1. Introduction

Memory and reflection are common components of language-agent systems. Generative Agents stores experiences, synthesizes reflections, and retrieves them during planning (Park et al., 2023); Reflexion uses verbal feedback in an episodic buffer (Shinn et al., 2023). Financial-agent systems such as FinMem and FinAgent likewise combine decision making with layered or diversified memory (Yu et al., 2023; Zhang et al., 2024). These architectures motivate a basic audit question: if an agent changes a binary action after a loss-memory prompt, did the retrieved outcome cause the change, or did the surrounding instruction do the work?

This paper addresses that question narrowly. It does not benchmark a trading system, seek alpha, or evaluate a live agent. It freezes a historical replay, queries each setup from a fresh model context under paired prompts, and records every prompt and response. The central result is negative for memory content but positive for prompt severity: in the tested configurations, retrieved negative versus balanced returns did not change actions, whereas warning language did.

2. Protocol and scope

Replay and decision rule

The frozen data are daily SPY OHLCV observations from 2018-01-01 through 2023-12-31. A candidate BUY setup occurs when 14-day RSI crosses from at least 40 to below 40 after 2021-01-01. Each setup's realized outcome is its ten-trading-day forward return. A record may enter retrieval only when its outcome date is on or before the later decision date; no unresolved forward outcome is supplied as memory. The frozen data SHA-256 is ``b58919ba5cf55a4529e96a0a0b90a2426cd4d6e79682636e02d1d3bc8ee8e768``.

The agent returns one of two actions: EXECUTE or ABORT. EXECUTE is paired with the frozen ten-day return; ABORT is assigned zero only for descriptive paired consequences. This is not a portfolio backtest: setups overlap, costs are absent, memories are teacher-forced, and the model does not interact with a market.

Experiment 1: low-pressure audit

Experiment 1 re-ran a corrected audit in one local Llama-3 8B build. It included 32 signal setups, of which 24 had at least one matured record. Four direct conditions separately varied a trauma frame, a neutral frame with the same losses, neutralized reflection prose, and action-neutral padded text. The 24 memory-eligible setups produced zero paired switches in every direct contrast; the four arms were all EXECUTE. An auxiliary label-by-number prompt factorial also produced only EXECUTE decisions. A later binding-stop diagnostic caused 24/24 ABORTs. It was a post-null harness check, not a pre-specified estimate of memory effect.

Experiment 2: severity, retrieval, and model follow-up

Before Experiment 2 inference, the analysis sample was fixed to 19 setups having at least two matured negative and one matured positive prior outcome in the same 200-day-SMA regime. Three local builds were queried from fresh contexts at temperature 0 and seed 20260714: Llama 3.2 3B Q4_K_M, Qwen 2.5 7B Q4_K_M, and Llama 3 8B Q4_0. Model digests are archived in the runtime manifest.

Each setup crossed three two-slot retrieval compositions: (i) the two worst prior negative outcomes; (ii) the worst negative plus the best positive outcome; and (iii) character-length-matched neutral padding without outcome facts. Each record gave date, regime, ten-day return, and maximum drawdown. Each composition crossed five fixed guidance levels: neutral, caution, strong caution, non-binding committee recommendation, and binding stop. The primary non-binding interval comprises neutral through discretionary recommendation. The full design contains $19 \times 3 \times 3 \times 5 = 855$ decisions.

The pre-execution plan specifies action rates and paired switch counts. Exact paired binomial p-values are reported only as fixed-sample summaries; the 19 setups are not a random sample from a population of markets, and temperature-zero calls are not independent human trials.

Experiment 3: generic-risk confirmatory control

Experiment 3 was planned and written after Experiment 2 had completed but before its own model calls. It uses the same 19 setups and three models, but includes no retrieval block and no memory, record, history, or outcome language. It compares generic neutral guidance with: "Current risk is material. Prefer ABORT unless the current state gives a concrete reason to execute." The $19 \times 3 \times 2$ design contains 114 decisions. This test asks whether generic caution alone is sufficient; it does not retroactively make Experiment 2 pre-registered.

3. Results

Experiment 1: no detected low-pressure memory effect

The corrected low-pressure audit did not reproduce the exploratory draft's 11 ABORTs in 34 setups. In the corrected run, the loss-trauma arm made 0/32 ABORTs, and all four direct conditions produced the same action on all 24 memory-eligible setups. Thus Experiment 1 provides no causal evidence that the tested negative-memory configuration changed action. It also provides no evidence of economic benefit or harm because there was no action selection to evaluate.

Experiment 2: warning severity, not return composition, changes action

Table 1 gives ABORT counts by model, warning severity, and retrieval composition. The strong pattern is severity sensitivity. Llama 3.2 and Llama 3 8B executed all setups under neutral and caution prompts, then aborted all setups under strong caution across negative-only, balanced, and padded-control contexts. Qwen aborted all negative-only and balanced setups beginning at caution. The binding stop caused 19/19 ABORTs for every model and retrieval condition.

Model	Neutral	Caution	Strong caution	Discretionary	Binding stop
Llama 3.2 3B	0/19 neg; 0/19 bal; 0/19 ctl	0/19 neg; 0/19 bal; 0/19 ctl	19/19 neg; 19/19 bal; 19/19 ctl	19/19 neg; 19/19 bal; 19/19 ctl	19/19 neg; 19/19 bal; 19/19 ctl
Qwen 2.5 7B	0/19 neg; 0/19 bal; 3/19 ctl	19/19 neg; 19/19 bal; 19/19 ctl	19/19 neg; 19/19 bal; 19/19 ctl	19/19 neg; 19/19 bal; 19/19 ctl	19/19 neg; 19/19 bal; 19/19 ctl
Llama 3 8B	0/19 neg; 0/19 bal; 0/19 ctl	0/19 neg; 0/19 bal; 0/19 ctl	19/19 neg; 19/19 bal; 19/19 ctl	0/19 neg; 0/19 bal; 11/19 ctl	19/19 neg; 19/19 bal; 19/19 ctl

Table 1. ABORT counts. 'neg', 'bal', and 'ctl' denote negative-only, balanced, and neutral-control retrieval, respectively. Each count has denominator 19.

The planned retrieval-content contrast is negative-only versus balanced. It yielded zero paired switches in every one of the 12 comparisons formed by three models and the four non-binding severity levels. In particular, replacing one severe historical loss with the highest-return positive outcome did not change a single action at fixed warning severity. This is no detected retrieval-valence effect, not proof of equivalence.

The neutral-control comparisons were less tidy and are therefore not used to claim a memory effect. Qwen made three more ABORTs under neutral padding than negative-only retrieval at neutral guidance (3/19 versus 0/19; exact paired $p = 0.25$). Llama 3 8B made 11 more ABORTs under neutral padding than negative-only retrieval under the discretionary committee wording (11/19 versus 0/19; $p = 0.00098$). Because the padding and guidance still referred to review or history, these irregularities show prompt-context sensitivity, not a valid content effect.

Several severity contrasts were complete 19/19 action flips. The planned exact fixed-sample p-value is $3.8e-6$ for a unidirectional 19-switch contrast, but the more useful fact is descriptive: a single warning phrase moved an entire tested action set. The paired action-return difference in those contrasts merely equals the mean return of the same 19 frozen setups; it is not

an estimate of selective trading skill and is deliberately not used as a headline result.

Experiment 3: generic warning wording is sufficient for the Llama switches

Table 2 removes all memory and history terms. Both Llama builds again changed all 19 actions from EXECUTE under generic neutral guidance to ABORT under generic strong caution. Qwen chose ABORT under both generic conditions, demonstrating a different baseline under this shorter prompt rather than a caution effect. Thus generic risk wording alone is sufficient for the Llama action shifts observed in Experiment 2.

Model	Generic neutral	Generic strong caution	Paired switches
Llama 3.2 3B	0/19 ABORT	19/19 ABORT	19 EXECUTE-to-ABORT
Qwen 2.5 7B	19/19 ABORT	19/19 ABORT	0 EXECUTE-to-ABORT
Llama 3 8B	0/19 ABORT	19/19 ABORT	19 EXECUTE-to-ABORT

Table 2. Experiment 3 generic-risk control. No prompt contains a retrieval block or historical/outcome language.

4. Interpretation

The evidence supports a limited causal statement: in these local-model replays, generic risk wording can change binary action, while the tested sign composition of retrieved return facts showed no detected action effect. It does not support the stronger statement that negative memory made the agent cautious, that it improved or degraded trading, or that the agent responds to emotional language rather than numerical magnitude.

This distinction matters because a system prompt can smuggle a conclusion into the decision layer. “Historical outcomes indicate material risk” is already a recommendation about what the agent should do. Experiment 3 shows that even the historical premise is unnecessary for the two Llama models once generic strong-caution wording is supplied. The right engineering conclusion is not that memory is useless. It is that memory evaluations need matched guidance controls before an action change can be credited to retrieved information.

5. Limitations and reproducibility

The study covers three local quantized builds, one asset, a single historical path, and 19 selected setups in the factorial follow-up. It is teacher-forced rather than live. Calls used fresh contexts, temperature 0, and a fixed seed, but this does not establish deterministic model semantics; two of 44 accidental duplicate calls produced different rationales while all 44 duplicate decisions agreed. The early run accidentally created 44 duplicate raw calls among the first 99 unique factorial cells. The immutable raw log is retained; the clean ledger selects the earliest valid response per cell by append order, contains exactly 855 unique Experiment 2 cells, and the reconciliation file documents this rule. This operational problem is a limitation, not a result.

Experiment 1 and the Experiment 2 follow-up plans were written before their respective calls, but the work was iteratively hardened after earlier audits; it was not prospectively registered. Experiment 3 was pre-specified only relative to its own calls. The reported exact paired p-values do not create population-level inference. Finally, a uniform action switch cannot demonstrate economic selection. The artifacts contain frozen data hashes, plans, runtime manifests, prompts, raw responses, clean ledgers, analysis code, and the overlap reconciliation.

References

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